TECHNICAL SUPERIORITY REPORT

CNT(Carbon Nano Tube) Conductive Tile

Nanotechnology-Based Technical Superiority Beyond Carbon Black

Superiority of CNT Technology Verified by Conductive Mechanism, Microstructure, and Official Test Data

DOOHA | 2026

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BACKGROUND

Limitations of Conventional Carbon Black ESD Tiles

High-Volume Carbon Black Loading to Secure Conductivity

Typically requires a large addition of 5–20 wt% or more to form conductive paths

Surface Dusting & Staining Issues

Surface contamination, aesthetic degradation, and work environment risks due to filler sloughing

Risk of Degradation in Mechanical Properties

Potential decrease in wear resistance, flexibility, and impact resistance due to high filler loading

Risk of Resistance Variations

Surface resistance may become unstable over temperature/humidity changes and wear progress

Industry Standard

5~20

wt%

Carbon Addition Range

→ The core of material improvement is forming "a more stable network with less amount"

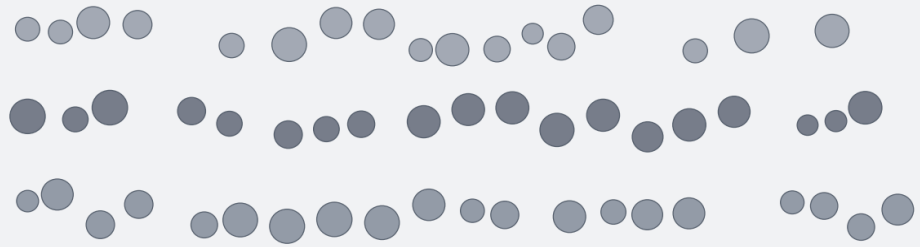
CORE PRINCIPLE

Why Different Structures Lead to Super Performance

Fundamental Differences in Conductive Network Formation Mechanism

Carbon Black (Conventional Carbon)

Spherical Particles · Point-Contact Method



- Spherical particle structure with diameters of tens to hundreds of nm
- 'Direct contact' network where current flows only when particles physically touch each other
- Must be densely filled over a large area to maintain contact points → High filler loading is inevitable

CNT (Carbon Nano Tube)

Fibrous Structure · Network-Bridging Method



- High aspect ratio fibers with diameters of a few to tens of nm and lengths thousands of times their diameter
- A single fiber strand 'bridges' across a wide area, forming a 3D network with even a trace amount
- Establishes continuous conductive paths with low addition level → Percolation Threshold significantly reduced

DATA 01 · LOADING wt%

Loading Level : Same Conductive Performance, Lower wt%

Carbon Black

5~20

wt%

Typical loading range required to secure target surface resistance

Up to 99% ↓

CNT (Carbon Nano Tube)

0.01~1

wt%

Loading range required to secure equivalent or superior conductive performance

Cause of Low Percolation Threshold: *The high-aspect-ratio fibrous structure of CNT forms a 'bridging network' rather than 'direct contact' between particles, completing continuous conductive paths with a significantly lower amount.*

Source: Comprehensive industry public technical data (OCSiAl TUBALL, Polymer Nanocentrum, etc.) — Actual formulation ratios may vary depending on tile structure, binder, and target resistance values.

Retention of Physical Properties & Surface Quality

Comparison Item	Carbon Black	CNT (Carbon Nanotube)
Mechanical Properties	Risk of decrease in strength, flexibility, and impact resistance due to high filler loading	Preservation of raw material inherent properties (wear resistance, flexibility) by maintaining low content
Surface Dusting · Contamination	Possibility of dusting and surface staining due to filler sloughing	Clean surface with low addition level, minimizing sloughing/dusting risks
Color Design Freedom	Constraints on color and aesthetic design due to high blackness	High design freedom for various colors and patterns through trace addition

Resistance Stability & Durability

Comparison Item	Carbon Black	CNT
Resistance Stability	Surface resistance may show deviations according to temperature/humidity changes and wear progress	Maintains stable resistance values even under environmental changes due to 3D network structure
Application Uniformity	Concerns of Hot-Spots in Chopped Carbon Fiber, etc. if minimum fiber length is not secured	Forms uniform conductive paths across various thicknesses and flooring structures
Long-Term Durability	Concerns of resistance increase and ESD performance degradation when filler is lost due to wear/delamination	Maintains performance even after repeated use and wear due to low content and stable bonding

VERIFICATION

DOOHA's Verification Method: Precision FESEM Analysis

DOOHA verifies the material microstructure through **precision analysis data from third-party authorized testing institutes**

Testing Institute	Korea Polymer Testing & Research Institute, KOPTRI (KOLAS Accredited)
Test No.	26-06106
Test Item	Morphology Analysis — FE-SEM
Test Conditions	Accelerating Voltage 5kV · Magnifications x20,000 / x50,000 / x100,000
Measurement Points	Repeated measurements at 5 points (Point 1~5) across sample surface
Test Period	2026.07.13 ~ 2026.07.20

Test Sample

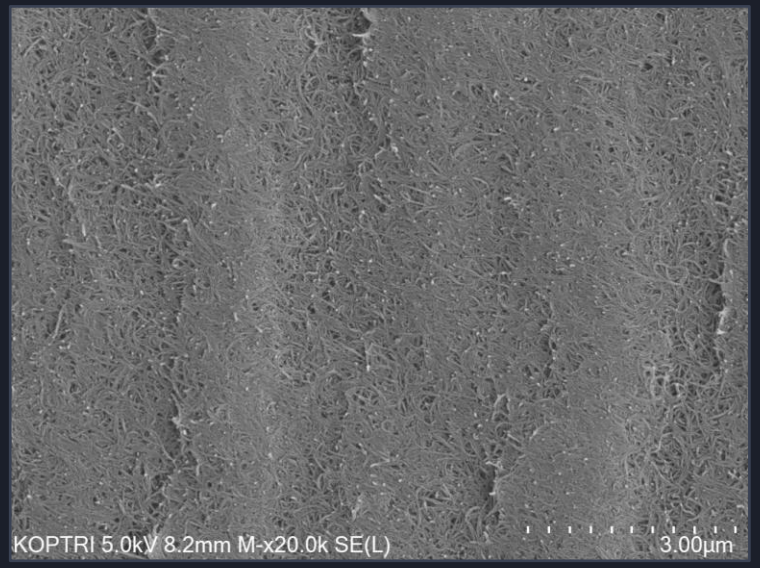


※ **Note:** The above sample (Carbon(TM-050), Applicant: TECHSENSE) is an authorized analysis case formally requested and tested by KOPTRI. DOOHA objectively verifies the microstructure of its own and benchmarked materials through the same third-party precision analysis method..

FESEM IMAGING

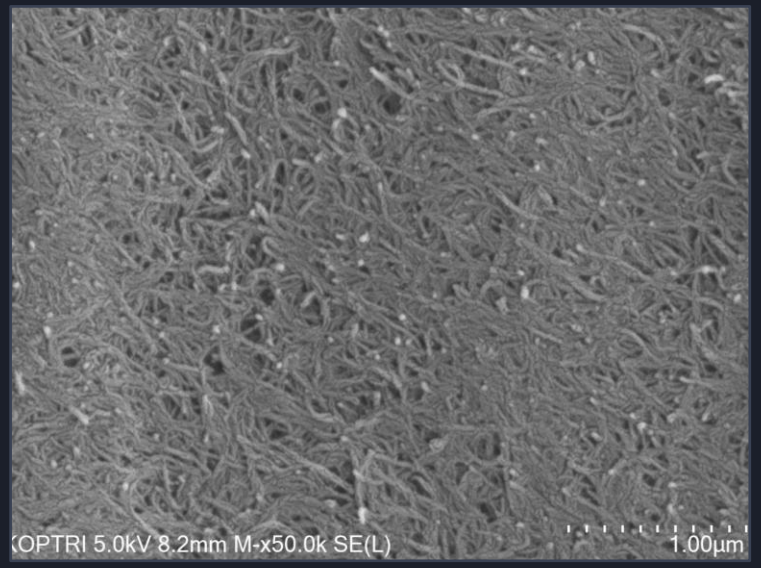
Microstructure Observation by Magnification

Koptri-26-00-09258-1 · Point 1 · Observation of the same point at 3 magnification levels



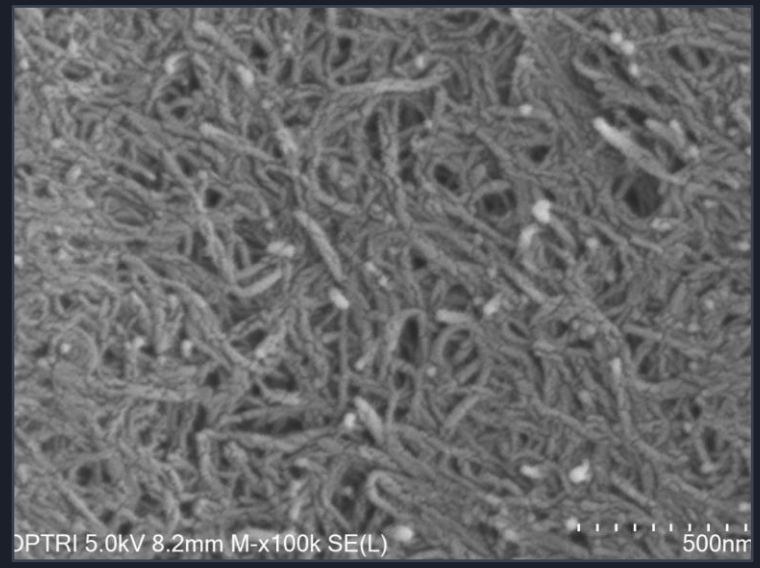
x20,000

3.00 µm scale



x50,000

1.00 µm scale



x100,000

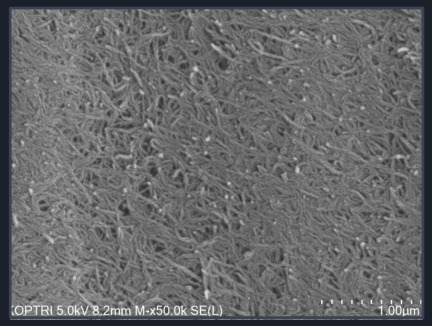
500 nm scale

As magnification increases, the 3D network structure formed by entangled fine fiber strands is clearly observed — microstructural evidence supporting why continuous conductive paths can be formed with small amounts.

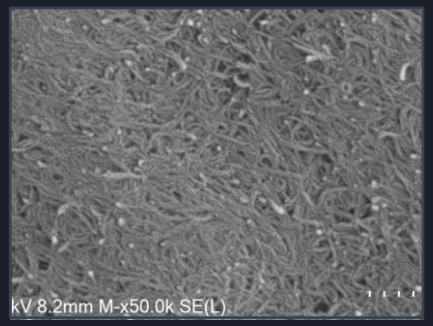
UNIFORMITY CHECK

5 Points Across Sample Surface — Confirmation of Consistent Microstructure

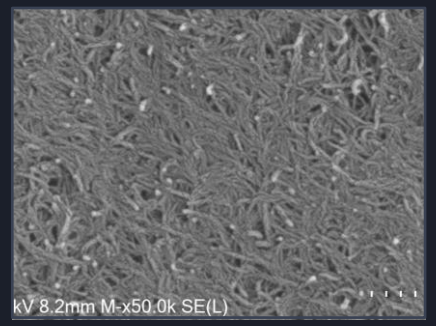
Based on x50,000 Magnification, Captured under identical conditions for Point 1~5



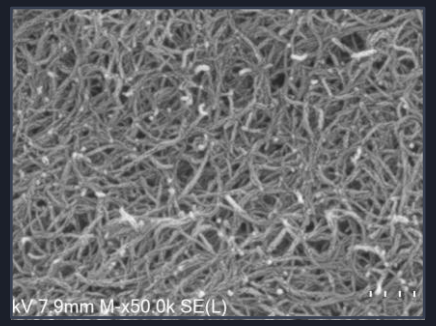
Point 1



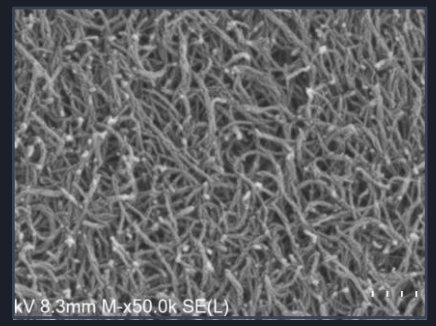
Point 2



Point 3



Point 4



Point 5

Similar fibrous network structures are repeatedly observed across all 5 points, supporting the microstructure uniformity over the entire material. Uniform network distribution is the key basis leading to stable and predictable surface resistance implementation across the tile.

SUMMARY

Comprehensive Comparison Table — CNT vs Carbon Black

Comparison Item	Carbon Black (Conventional Carbon)	CNT (DOOHA CNT Coating)
Conductive Mechanism	Direct contact between spherical particles	3D network of high-aspect-ratio fibers
Loading Level	Approx. 5~20 wt%	Approx. 0.01~1 wt%
Mechanical Properties	Risk of degradation due to high filler loading	Preservation of raw material properties with low content
Surface Quality	Dusting and staining possible	Clean surface, high color freedom
Resistance Stability	Deviations possible due to environmental changes	Deviations minimized through stable network
Verification Method	-	Authorized FESEM analysis via KOPTRI (KOLAS), etc.

STANDARDS

Global ESD Standards Compliance Capability

CNT coated conductive tiles are engineered targeting electrical properties capable of complying with major international and domestic ESD/conductive flooring standard systems.



ANSI/ESD S20.20

Representative international standard for Electrostatic Discharge (ESD) control programs



IEC 61340-4-1

International standard for measuring electrical properties (resistance) of floor coverings



ASTM F150

Standard test method for electrical resistance of conductive resilient flooring (USA)



KS M 3802

Korean Industrial Standard for floor coverings — Thickness tolerance, heat shrinkage, etc.

CONCLUSION

Next-Generation ESD Solution Opened by CNT

Minimum Loading, Maximum Conductive Performance

Achieves equivalent or superior conductive performance with overwhelmingly lower dosage compared to conventional carbon at a trace level of 0.01–1 wt%

High-Durability ESD Tile Without Degradation of Physical Properties

Preserves inherent wear resistance and flexibility of raw material by maintaining low content, minimizing surface dusting and staining risks

Reliability Based on Authorized Institute Verification

Proven by data through precision FESEM analysis from third-party authorized testing institutes like KOPTRI

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